(Please write your Enrollment Number) Enrollment No. 035

End-Term Examination
(CBCS)(SUBJECTIVE TYPE)(OffLine)
Course Name:<B. Tech CSE/IT/MAE/ECE/DMAM/ECE-AI >, Semester:<3rd>
(November-December, 2023)

Subject Code: BIT 201	Subject: DBMS

Time :3 Hours Maximum Marks :60

Note:Q. 1 is compulsory. Attempt one question each from the Units I, II, III & IV.

Q1		(4*5=20)	
	a) Define database management system and what are it's various applications? What are the different anomalies in designing a database?		
	b) What is atomicity of a transaction? Explain with example.		
	c) What are the various functionalities of a typical database management system?		
	d) What do you understand by conflict serializability? Explain with the help of an example.		
	UNIT-I		
Q2	List and explain various data models used for database design.	(10)	
Q3	What is Entity set and also define Relationship set. List and explain the symbols used to draw ER Diagram.	(10)	
	UNIT-II		
Q4	Write SQL Queries for following set of tables: EMPLOYEE (EmpNo, Name, DoB, Address, Gender, Salary, DNumber) DEPARTMENT (DNumber, Dname, ManagerEmpNo, MnagerStartDate). i) Display the Age of 'male 'employees. ii) Display all employees in Department named 'Marketing'. iii) Display the name of highest salary paid 'female ' employee. iv) Which employee is oldest manger in company? v) Display the name of department of the employee 'SMITH'.	(10)	

Q5	Create the E-R diagram for a university database and stepwise convert it into a relational database using the mapping algorithm.	(10)	
	UNIT-III		
Q6	Normalize following relation up to 3NF: Bank(acno, cust_name, ac_type, bal, int_rate, cust_city, branchId, branch_nm, br_city)	(10)	
Q7	What do you understand by functional dependencies? Given a relation R(A, B, C, D) and Functional Dependency set FD = { AB → CD, B → C }, determine whether the given R is in 2NF? If not convert it into 2 NF.	(10)	
	UNIT-IV		
Q8	List and explain the various issues while transactions are running concurrently in DBMS.	(10)	
Q9	Explain Concurrency control with locking methods.	(10)	